

AHMED MOHAMED SOBHY MOSTAFA

+201142414557 | www.linkedin.com/in/ahmedsobhyyy | ahmedshobhy2007@gmail.com | Giza, Egypt
<https://ahmedsobhy78.github.io/Ahmed-Sobhy-Portfolio/>

Mechatronics and Robotics Engineering student with hands-on experience in electric systems, robotics and automotive engineering. Leader of the Electrical Team at Spark X Racing Team, with a strong passion for electric vehicles, electrical components, and control systems. Skilled in CAD, diagnostics, and teamwork, with strong leadership, communication, and problem-solving abilities.

EDUCATION

Helwan National University 2024 - 2028
B.Sc. in Mechatronics and Robotics Engineering

EXPERIENCE

Intern Aug – Sep 2025
El Sewedy Industries

- Successfully completed the Summer Internship Program at El Sewedy Industries Group.
 - Learned about industrial electrical systems, manufacturing processes, and production line operations.
 - Observed maintenance procedures, industrial safety standards, and quality control systems inside factory environments.
 - Gained exposure to electrical components, automation systems, and industrial workflow management.
 - Participated in technical visits to ASF (Arab Steel Fabrication), EMAS (Egyptian Manufacturing and Advanced Systems), and EGYLUX under El Sewedy Industries collaboration.
 - Developed a better understanding of fabrication systems, electrical manufacturing, and industrial engineering applications, Observed maintenance tasks such as AC evaporator replacement and timing belt installation.

Smart Systems Workshop — KUKA Official System Oct - 2025

- Learned how to operate and control KUKA industrial robots in automation applications.
- Performed robotic movement tasks and gained hands-on experience with robot programming concepts.
- Developed understanding of robotic manufacturing systems, industrial automation, and embedded control.
- Learned industrial safety procedures and operational standards for working with robotic systems.

Automotive Intern

Honda Clinic – Car Maintenance Center Jul 2026 | 1 Week

- Successfully completed a one-week summer internship at Honda Clinic Car Maintenance Center.
- Gained hands-on experience in automotive maintenance and vehicle inspection procedures.
- Observed maintenance and troubleshooting of engine, braking, suspension, and electrical systems.
- Learned preventive maintenance techniques and workshop safety standards.
- Developed practical knowledge of automotive diagnostics and professional service operations.

Mechanical Intern

Talaat Moustafa Group (TMG) – Atrium & Square Projects

Jul 2026 | 2 Weeks

- Completed a two-week mechanical internship at Talaat Moustafa Group across Atrium and Square projects.
- Gained practical experience in HVAC, firefighting, plumbing, and drainage systems.
- Observed installation, testing, and commissioning of mechanical systems.
- Learned about MEP coordination, technical drawings, and construction site engineering practices.
- Followed engineering standards and site safety procedures during project execution.

Cairo International Airport – Engineering Training

Aug – Sep 2026 | 1 Month

- Gained practical exposure to HVAC systems, chillers, diesel generators, and general electrical power systems supporting airport operations.
- Learned how electrical, mechanical, and backup power systems are interconnected and coordinated to maintain continuous power and reliable airport operations.
- Explored the airport's engineering infrastructure, including rooftop HVAC equipment, fire protection systems, and critical operational facilities.
- Understood the importance of reliable infrastructure in airports and other critical facilities, where power interruptions or operational delays can significantly impact safety, efficiency, and travel operations.

Electrical Team Leader

Feb 2025 - Present

Spark X Racing Team

- Leading the Electrical Team in developing and improving electrical systems for Formula Student projects.
- Working on electrical components, wiring systems, and vehicle electrical integration.
- Coordinating team tasks and supporting technical problem-solving during projects and competitions.
- Contributing to innovation and development related to electric vehicle technologies.

PROJECTS

- **Modeling and Embedded System Design (Formula Student)** – Contributed to the modeling and embedded systems development for Formula Student applications within Spark X Racing Team.

Worked on integrating embedded systems and electrical components to support vehicle functionality and performance. Assisted in system design using LTspice, simulation, and testing processes to ensure efficient operation and reliability.

Gained practical experience in automotive embedded systems, electrical integration, and engineering problem-solving within a competitive environment

- **Engineering Project Showcase Participant (RC BOAT)**

Designed and showcased an RC Boat project. Presented it to prospective students, faculty members, and university leadership, promoting hands-on engineering and innovation. Received a Certificate of Appreciation for my contribution.

- **Digital Clock Using ICs & Proteus (Electronic Project)**

Designed and simulated a fully functional digital clock circuit using logic ICs on Proteus.

Implemented timing and counting operations through integrated circuits and digital electronics concepts.

Worked on circuit design, debugging, and simulation to ensure accurate time display and stable operation.

Strengthened understanding of digital electronics, timing systems, and circuit analysis, maze-solving logic and sensor integration

- **Uninterruptible Power Supply (UPS)**

Designed and developed a UPS system integrating a custom-built inverter, transformer, relay, and voltage regulation circuit.

Designed the inverter circuit from scratch using Arduino Nano, IRFZ44N MOSFETs, and 2N2222 transistors for switching and control.

Simulated and tested the complete circuit using Proteus to analyze switching behavior and circuit performance.

Applications: Backup power systems for hospitals, data centers, telecommunications systems, industrial equipment, and critical electronic devices.

- **Cam Design and Motion Analysis (CAD)**

Designed mechanical cam profiles and analyzed follower motion characteristics using engineering principles.

Calculated and plotted displacement, velocity, and acceleration equations using Microsoft Excel.

Performed motion analysis to study cam performance and optimize mechanical movement behavior.

Improved analytical and problem-solving skills in mechanism design and kinematic analysis.

- **Quick Return Mechanism (QRM) Shaping Machine Model**

Designed and built a mechanical model demonstrating the Quick Return Mechanism used in shaping machines, for educational demonstration.

TECHNICAL SKILLS

- **Robotics & Control:** Arduino, ESP32, Embedded Systems, Robotics Control
- **Design & Simulation:** CAD (Autodesk Inventor), OptimumKinematics, Simulink Modeling
- **Automotive:** Automotive Electrical Systems, Diagnostics & Troubleshooting.
- **Other:** Electronics Prototyping, Social Media Management, Graphic Design

SOFT SKILLS

- **Leadership & Team Management**
- **Problem-Solving & Analytical Thinking**
- **Working Under Pressure**
- **Communication & Collaboration**
- **Public Speaking & Presentation**

LANGUAGES

- **Arabic: Native**
- **English: Fluent (C1)**